

Yi Chen

+1 984-259-9266 | yichenyc@umich.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Michigan – Ann Arbor

M.A. in Business Administration – Quantitative Marketing

• Ph.D. program (master out for industry), GPA: 4.0/4.0

• **Relevant Courses:** Reinforcement Learning Theory, Machine Learning, Econometric Analysis, AI Foundations

Duke University

M.A. in Economics, GPA: 3.9/4.0

• **Relevant Courses:** Causal Inference, Probabilistic Machine Learning, Bayesian Modeling, Natural Language Processing

Hong Kong University of Science and Technology

B.S. in Economics and Finance, GPA: 3.8/4.3 (Top 5%)

PROFESSIONAL EXPERIENCE

Jiangsu Zucai E-Business Co., Ltd

Research Collaborator

Shanghai, remote from Ann Arbor, MI

Oct 2024 – present

Research paper: Suppliers' Responses to Recommendations on B2B E-Commerce Platforms [Marketing Science Conference 2026, Theory + Practice in Marketing 2026, Platform Strategy Research Symposium 2026]

- Evaluated the impact of personalized supplier recommendations on bidding participation and downstream marketplace outcomes using 6M+ recommendation records and 20M+ bidding records from a large B2B e-commerce platform.
- Built an end-to-end causal study, with a data pipeline in Python and MySQL, applying Coarsened Exact Matching, Propensity Score Matching, and Double Machine Learning with Random Forest nuisance models to estimate direct, spillover, and heterogeneous treatment effects across supplier segments.
- Quantified a 1.4 pp direct uplift and 0.7 pp spillover uplift in bidding participation, identified materially different responses across niche and mainstream suppliers, and proposed optimizing recommendation objectives toward bid success and downstream conversion rather than bidding volume alone.

DeepMannifold

Machine Learning Engineer Intern

New York, NY

Jun 2026 – Aug 2026

- Contributed to the development of a real-time Smart Bidding system for a start-up client that maximizes bidding pace and ROI while achieving high throughput, low latency, and real-time bidding at scale.
- Designed a pCTCVR-based bid-pricing module that converted conversion value into impression-level expected value and bid prices, using bid-landscape modeling and historical auction replay to evaluate win/loss outcomes, market thresholds, paid price, and frequency caps.
- Implemented LinUCB cold-start exploration and budget-aware PPO/GRPO bidding policies using ESMM outputs and pacing state to learn first-price bid-shading ratios
- Added multi-period pacing, safety guardrails, staged rollout, schema validation, monitoring, rollback/fallback paths, MLflow-tracked offline replay, and simulation-based evaluation, supporting a controlled advertiser beta that reduced CPA by ~8% and pacing error by ~25% while maintaining conversion volume.

PROJECT EXPERIENCE

Predicting Incremental Effects of Headlines in A/B Tests

May 2026 – present

- Built a prediction framework using 30K+ historical A/B tests to forecast the incremental effect of news headlines, enabling editorial teams to prioritize which experiments to run without testing every candidate.
- Applied Bayesian models (such as Bayesian Additive Regression Trees, Bayesian Neural Networks) on text embeddings to estimate treatment effects with calibrated uncertainty, using JAX for accelerated computation on cloud computing clusters.
- Improved prediction reliability using a posterior recalibration technique, reducing prediction error by 15% and increasing confidence interval coverage from 89% to 94%.

Multi-Layered Customer Intelligence Platform

Feb 2026 – May 2026

- Designed a 3-layer embedding engine to transform bank customers' data into insightful segmentations for product recommendation, combining transaction sequence modeling (Word2Vec), self-supervised contrastive learning (SimCLR), and graph-based representations (GraphSAGE) to generate 128-dim customer embeddings for downstream tasks.
- Built and maintained a self-supervised learning pipeline using SimCLR, reducing dependence on labeled data while preserving representation quality for segmentation and recommendation.
- Developed and evaluated a Deep Interest Network (DIN) with attention mechanisms for personalized product recommendation.

Intent-Aware Graph Retrieval for Long-Term Conversational Memory

Feb 2026 – May 2026

- Designed an external memory system for long-term LLM dialogues to address information fragmentation and multi-hop reasoning in conversational AI.
- Proposed a novel graph-based retrieval algorithm that improves upon a state-of-the-art memory system (MAGMA) by better distinguishing relevant from irrelevant retrieved information.
- Benchmarked the system using an LLM-as-judge evaluation protocol, improving retrieval accuracy by 1.4 pp on average, particularly on open-domain (+10.0 pp) and multi-hop (+5.7 pp) questions.

SKILLS

Programming Skills SQL/MySQL, Python (Pandas, NumPy, Scikit-Learn, PyTorch), R, Spark, Hadoop

Analytical Skills Causal Inference, A/B Testing, Experimental Design, Time Series Analysis, Supervised/Unsupervised Learning, Deep Learning, Natural Language Processing